Course Title: Mathematics-I for CSE Stream/ECE Stream **Semester:** I

Course Code: 25BEPHY101/25BEELY101 **Credits:** 4

QUESTION BANK

MODULE-1: Differential Calculus

1. Prove that with usual notation $\cot\phi = \frac{1}{r}\frac{dr}{d\theta}$

2. Find the angle between the radius vector and the tangent to the curve:

i. $r = a(1 - \cos\theta)$

ii. $r^2 \cos 2\theta = a^2$

iii. $r^m = a^m(\cos m\theta + \sin m\theta)$

iv. $\frac{2a}{r} = 1 - \cos\theta$ at $\theta = \frac{2\pi}{3}$

v. $rcos^2\left(\frac{\theta}{2}\right) = a$ at $\theta = \frac{2\pi}{3}$

3. Show that the following pairs of curves intersect each other orthogonally:

i. $r = a(1 + \cos\theta)$ and $r = b(1 - \cos\theta)$

ii. $r = a(1 + \sin\theta)$ and $r = a(1 - \sin\theta)$

iii. $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$

iv. $r^2 \sin 2\theta = a^2$ and $r^2 \cos 2\theta = b^2$

v. $r = ae^{\theta}$ and $re^{\theta} = b$

4. Find the angle of intersection of the following pairs of curves:

i. $r = \sin\theta + \cos\theta$ and $r = 2\sin\theta$

ii. $r = alog\theta$ and $r = \frac{a}{log\theta}$

iii. $r = a(1 - \cos\theta)$ and $r = 2a\cos\theta$

iv. $r = a(1 + \cos\theta)$ and $r^2 = a^2\cos 2\theta$

v. $r = a\theta$ and $r = \frac{a}{\theta}$

5. Find the pedal equation of the following curves:

i. $\frac{2a}{r} = (1 + \cos\theta)$

ii. $r(1 - \cos\theta) = 2a$

iii. $r^n = a^n \cos n\theta$

iv. $r^m = a^m(\cos m\theta + \sin m\theta)$

v. $\frac{l}{r} = 1 + e\cos\theta$

Radius of Curvature:

1. Find the radius of curvature $x^3 + y^3 = 3axy$ at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on it.
2. Find the radius of curvature at any point $P(x, y)$ on the parabola $y^2 = 4ax$.
3. Find the radius of curvature of the curve $y = a\log\left(\sec\left(\frac{x}{a}\right)\right)$ at any point (x, y).
4. Find the radius of curvature of the curve $x^2y = a(x^2 + y^2)$ at the point $(-2a, 2a)$.
5. Find the radius of curvature of the curve $a^2y = x^3 - a^3$ at the point where the curve cuts X-axis or at the point $(a, 0)$.
6. Find the radius of curvature of the curve $r^n = a^n \cos n\theta$.
7. Show that for the curve $r(1 - \cos\theta) = 2a$, ρ^2 varies as r^3.
8. Prove that $\frac{\rho^2}{r}$ is a constant for the cardioid $r = a(1 + \cos\theta)$ where ρ is the radius of curvature.
9. For the curve $r = ae^{\theta\cot\alpha}$, prove that $\frac{\rho}{r}$ is constant.

MODULE-2:

Series Expansion and Multivariable Calculus

Taylor's series and Maclaurin's series

1. Expand $f(x) = tanx$ in powers of $\left(x - \frac{\pi}{4}\right)$ up to third degree terms.
2. Expand $f(x) = sin(e^x - 1)$ in powers of x up to the terms containing x^4.
3. Obtain the Maclaurin's expansion of $sin^{-1}x$ up to the term containing x^3.
4. Expand e^{sinx} using Maclaurin's theorem up to the terms containing x^4.
5. Expand $\log(1 + e^x)$ using Maclaurin's theorem up to the terms containing x^4.
6. Using Maclaurin's series expand $f(x) = log(cosx)$ up to the term containing x^4.
7. Using Maclaurin's series expand $log(secx)$ up to sixth degree terms.
8. Expand $\log(1 + \sin x)$ using Maclaurin's series up to the term containing x^4.
9. Using Maclaurin's series expand $f(x) = tanx$ up to the term containing x^5
10. Using Maclaurin's theorem Prove that $\sqrt{1 + \sin 2x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} \mp \cdots$.
11. Expand $\log(1 + \cos x)$ using Maclaurin's series up to the term containing x^4.

Indeterminate forms

1. $\lim_{x\to 0} \frac{\sin x - x}{\tan^3 x}$	**2.** $\lim_{x\to \frac{\pi}{2}} \frac{\log(cosx)}{tanx}$
3. $\lim_{x\to \frac{\pi}{2}} [\sec x - \tan x]$	**4.** $\lim_{x\to 1} x^{\left(\frac{1}{1-x}\right)}$
5. $\lim_{x\to 0} \left(\frac{a^x + b^x}{2}\right)^{\frac{1}{x}}$	6. $\lim_{x\to 0} (x)^{sinx}$
7. $\lim_{x\to 0} (cotx)^{tanx}$	**8.** $\lim_{x\to 0} \left(\frac{a^x + b^x + c^x}{3}\right)^{\frac{1}{x}}$
9. $\lim_{x\to 0} \left(\frac{1}{x}\right)^{2sinx}$	**10.** $\lim_{x\to 0} \left(\frac{2^x + 3^x + 4^x}{3}\right)^{\frac{1}{x}}$
11. $\lim_{x\to 0} (cosx)^{\frac{1}{x^2}}$	**12.** $\lim_{x\to \frac{\pi}{2}} \frac{1-(cosx)}{xlog(1+x)}$
13. $\lim_{x\to 0} \left(\frac{sinx}{x}\right)^{\frac{1}{x^2}}$	14. $\lim_{x\to 0} \left(\frac{a^x + b^x + c^x + d^x}{4}\right)^{\frac{1}{x}}$

15. $\lim_{x\to 0}\left(\frac{tanx}{x}\right)^{\frac{1}{x}}$	16. $\lim_{x\to\frac{\pi}{2}}(sinx)^{tanx}$

Total derivative

Find the Total derivative of:

1. $z = x\,y^2 + x^2\,y$ where $x = at,\ y = 2at$.
2. $u = xy + yz + zx$ where $x = t\cos t,\ y = t\sin t,\ z = t,\ at\ t = \frac{\pi}{4}$.
3. $u = x^2 + y^2 - z^2$ where $x = e^t, y = e^t\cosh t,\ z = e^t\sinh t$.

Chain rule:

1. If $z = x^2 + y^2$ where $x = e^u\sin v,\ y = e^u\cos v$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ as a composite function and verify the results by direct substitution.
2. If $u = f\left(\frac{x}{y},\frac{y}{z},\frac{z}{x}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.
3. If $u = f(x-y, y-z, z-x)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.
4. If $u = f(2x-3y, 3y-4z, 4z-2x)$, show that $6\frac{\partial u}{\partial x} + 4\frac{\partial u}{\partial y} + 3\frac{\partial u}{\partial z} = 0$.

Jacobian:

1. If $u = x+y+z,\ v = y+z\ and\ z = uvw$, find the value of $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.
2. If $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$, Show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$.
3. If $u = \frac{2yz}{x}, v = \frac{3zx}{y}, w = \frac{4xy}{z}$, find the value of $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.
4. Find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ where $u = x^2+y^2+z^2, v = xy+yz+zx,\ w = x+y+z$.
5. If $u = x+3y^2-z^3, v = 4x^2yz,\ w = 2z^2-xy$ find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ at (1, -1, 0).
6. If $u = xy^2,\ v = yz^2,\ w = zx^2$, find the value of $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.
7. If $u = \frac{x}{y-z},\ v = \frac{y}{z-x},\ w = \frac{z}{x-y}$, find the value of $\frac{\partial(x,y,z)}{\partial(u,v,w)}$.